K ZAYD AHMED

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SUMMARY

GenAI engineer who builds and deploys **autonomous agents** and **RAG systems** in production. At Capgemini, designed a multi agent application on LangChain, LangGraph, and AutoGen that replaced a fully manual triage process for 14 application teams, with evaluation, safety guardrails, and monitoring in production. Outside work, builds agent evaluation frameworks and self verifying agents that resolve hallucinations.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Go
Agent Frameworks: LangChain, LangGraph, AutoGen, Model Context Protocol (MCP), Tool Calling, Orchestration
LLMs: Large Language Models (LLMs), Azure OpenAI, Anthropic Claude, Gemini, OpenRouter, Prompt Engineering
RAG & Retrieval: Retrieval Augmented Generation (RAG), Vector Databases (Milvus, FAISS), Embedding Models
Evaluation & Safety: Agent Evaluation, LLM as Judge, Guardrails, Hallucination Mitigation, Human in the Loop
NLP: Semantic Search, Sentence Transformers, Cross Encoder Reranking, spaCy, nDCG, MRR
Engineering: FastAPI, REST APIs, Docker, Git, GitHub Actions (CI/CD), pytest, Azure, PostgreSQL, Redis

PROFESSIONAL EXPERIENCE

Software Engineer (Analyst A4), Capgemini | Bengaluru, Karnataka — July 2025 – Present

- Built and deployed a **production multi agent system** in Python and FastAPI, a 9 node LangGraph state machine with AutoGen agents on Azure OpenAI, replacing a fully manual triage process for **14 application teams**
- Developed a **RAG pipeline** on a Milvus vector database with embedding based semantic search across internal business units, and applied Model Context Protocol (MCP) concepts to agent tool calls
- Implemented **safety guardrails** on every LLM output (JSON schema validation, a rule override, and a confidence floor) so hallucinated or costly answers escalate to a human reviewer instead of reaching users
- Evaluated the embedding retrieval layer against production data and replaced it with **235 deterministic rules**, cutting LLM inference calls and making every routing decision auditable
- Built **two autonomous agents**, an Inbox Triage Agent and an Incident Orchestration Agent, on a multi agent platform across 3 PoCs, automating alert triage worth **about two people of manual work**
- Implemented **monitoring and logging** of agent behavior through a dashboard of status, cost, and latency per run, and fixed a live concurrency bug with an atomic compare and swap that ended duplicate records under retries
- Backed the system with **303 unit, integration, and end to end tests** across 52 files in CI, demoed it to the client, and onboarded application teams onto it

PROJECTS

Tripwire, Evaluation and Safety Framework for a Tool Calling Agent — GitHub
Python, Pydantic, pytest, GitHub Actions, Gemini API

- Built an **agent evaluation framework** with 60 golden cases (15 adversarial) that assert tool call order, arguments, forbidden calls, and cost budgets, gated in GitHub Actions with **237 offline tests**
- Added an LLM as Judge with Cohen's kappa calibration, a record and replay gateway for deterministic CI, and per call tracing of latency, tokens, and cost that **caught a hidden Gemini 3 token billing gap**

Reasoning Agent Selfcheck, Autonomous Agent that Resolves Its Own Hallucinations — GitHub
Python, Anthropic API (Claude), Gemini API, OpenRouter, unittest

- Built a **planner, executor, and verifier agent** where the LLM emits expressions an AST allowlisted evaluator recomputes, and a **six check verifier** rejects mismatches, returning a stated failure instead of a guess
- Wrote a provider agnostic LLM interface across Claude, Gemini, and OpenRouter with retry, backoff, and JSON mode, backed by **217 offline tests** and prompt contracts for four prompts

Braid, Hybrid RAG Engine with Semantic Search and NLP Reranking — GitHub
Python, FAISS, OpenSearch, Neo4j, Sentence Transformers, spaCy, Docker

- Fused BM25, FAISS dense search, and Neo4j graph traversal with a **fine tuned cross encoder reranker**, serving queries at p50 0.99 s, and scored nDCG and MRR over **180 labeled queries** with bootstrap intervals

EDUCATION

B.Tech, Computer Science and Engineering, CMR College of Engineering and Technology — July 2025
GPA: 7.83

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate (AZ204) | Credential ID B7A2F586F1CD56DB